Real-Time AI Video Monitoring for Electoral Integrity in Armenia's 2026 Parliamentary Elections

Executive Summary

Recent deployments in India (Kerala, Bihar, several states using AI-enabled CCTV and webcasting) and commercial platforms like VMukti demonstrate that large-scale AI video monitoring of polling stations is technically feasible and already used by election commissions, mainly for queue management, security, and post-hoc evidence review rather than real-time legal qualification of violations. Academic work on automated voter counting, crowd analytics, and election forensics further confirms that modern object detection and tracking algorithms can reliably detect people, estimate density, and flag anomalies, provided camera placement and data quality are adequate. Armenia's Electoral Code already mandates full-time video recording and centralized webcasting from polling stations for National Assembly elections, which greatly lowers the organisational barrier for building an AI layer on top of the existing camera infrastructure, but also constrains what can legally be analysed (especially secrecy of the vote).[1][2] [3][4] [5][6] [7][8] [^9]

For the 2026 nationwide deployment across approximately 1800 polling stations, a cloud-native architecture based on GPU-accelerated video analytics (e.g., NVIDIA DeepStream + TensorRT running YOLO-family detectors with DeepSORT/FeatureSORT tracking) and horizontal scaling across multiple nodes is technically viable if per-stream frame rates are kept modest (e.g., 5–10 fps analytics while archiving full video), and models are pruned/quantized. However, the system cannot fully replace trained human observers for nuanced legal assessment of electoral violations; instead, the realistic and legally safer role is a central early-warning system that detects visual risk patterns (queue congestion, multiple entries into booths, repeated escorting of voters, crowding, aggressive behaviour) and escalates clips to a rapid-response legal and observer network.[10][11] [12][13] [^14]

Overall, from a purely technical perspective, such a system is feasible for Armenia's scale and timeline if (a) legal constraints on data processing are carefully respected, (b) the scope focuses on pattern-based risk alerts rather than automated accusation of specific individuals, and (c) there is sufficient investment in pre-election camera calibration, pipeline optimisation, and integration with human observers and lawyers.

1. Global Market and Academic Precedents

1.1 Governmental deployments and vendor solutions

Kerala (India) – AI queue monitoring and 100% webcasting
Kerala's 2026 Assembly elections are using an AI-powered video monitoring system to count voters in queues outside polling stations and automatically flag queues longer than 10 people so that staff can be dispatched to reduce waiting times. Kannur district, for example, has deployed 100% webcasting and camera surveillance inside and outside all 2177 polling stations, with feeds monitored in control rooms for transparency and bogus-voting prevention.[15][16] [17][18] [^1]

Key characteristics:

- Focus: Queue length estimation, congestion alerts, and general deterrence, not fine-grained fraud classification.
- Architecture: PTZ/IP cameras streaming to control rooms with humans validating AI alerts.[^1]
- Success/limitations: Improves responsiveness to congestion and offers large-scale visibility but raises privacy concerns and relies on human validation to avoid false alarms. [15][1]

Election Commission of India – AI-enabled CCTV surveillance
The Election Commission of India has piloted AI-enabled CCTV surveillance in nearly all polling stations of at least one state, using Wi-Fi FHD cameras whose streams are monitored at multiple levels (returning officer, district, chief electoral officer, national HQ). The system generates "intelligent alerts" from cloud-based analytics to flag potential malpractice or disruptive events.[19][6]

- Focus: Security, crowd disturbances, and evidentiary recording around EVMs and entrances; secrecy is preserved by camera placement.[^19]
- Algorithms: Details are not public, but typical commercial stacks use object detection (persons, crowds), loitering/anomaly detection, and region-based alarms.

VMukti – commercial election surveillance platform
VMukti Solutions markets itself as a leading provider for election webcasting and AI video analytics in India, with cloud VMS/VAS and 4G cloud CCTV at polling centres.[2][20] [21][5] [^22]

- Capabilities: Multi-booth webcasting, operator dashboards, AI models for crowd monitoring, tampering detection, and sometimes vote-counting or OCR on EVM screens. [22][7]
- Scale: Hundreds to thousands of cameras in national and state elections.[20][5]
- Failure modes: Network instability (4G), variations in lighting and camera angle that degrade analytics, and the need for substantial human supervision.

Bihar, India – OCR-based vote count verification
The Bihar State Election Commission used Staqu's JARVIS video analytics at counting centres to read EVM screen outputs via OCR and cross-check them against official digital records, raising alerts when discrepancies appeared.[^7]

- Algorithms: OCR/text recognition plus structured comparison; no identity-level monitoring of voters.
- Success: Demonstrated AI can perform sensitive cross-checks reliably at scale, but only on clear, structured visual targets (EVM screens) rather than complex crowd behaviour.

Armenia – nationwide web cameras in 2017
During Armenia's 2017 parliamentary and Yerevan City Council elections, the government, supported by EU funding, installed around 1500 cameras in polling stations to provide live streams accessible on the internet and for later complaint support. Civil society organisations monitored all camera feeds in dedicated centres to identify irregularities (e.g., multiple voters in booths, stuffing).[23][24] [^25]

- Architecture: Mostly passive streaming; no automated analytics layer.[^24]
- Lessons: Large-scale live streaming in Armenian polling stations is administratively possible, but manual monitoring is labour-intensive and many violations still occurred.[26][23]

1.2 Academic work on AI for polling-place monitoring

Automated voter counting and queue estimation
A 2024 paper from Bangladesh proposes an automated voter counting system using OpenCV, CVZone, and the MOG2 background-subtraction algorithm, aiming to benchmark accuracy, efficiency, and speed against manual counting. Crowd congestion monitoring research (e.g., fusing CCTV video with spatial floor-plan information) uses deep learning detection, Kalman filtering, and mobility graphs to derive per-area density and flows from fixed cameras.[3][8]

- Algorithms: Background subtraction (MOG2), YOLO-family detectors, Kalman-filter tracking, and crowd-density estimation over spatial grids.[8][3]
- Reported performance: Good density and count estimates in structured scenes, but sensitive to occlusion, camera height, and lighting.[^8]

Election forensics and anomaly detection
Machine-learning-based election forensics uses tabular results rather than video, training classifiers on vote/turnout distributions to detect suspicious polling stations. This is complementary rather than directly computer-vision-based but shows AI can pinpoint where to look more closely.[^4]

Face-based fraud detection prototypes
Multiple academic projects propose facial-recognition-based systems to prevent duplicate voting or impersonation, often using OpenCV + deep face recognition libraries or face-api.js to compare each voter with a pre-enrolled database.[27][28] [29][30] [^31]

- Algorithms: CNN-based face recognition (e.g., DeepFace, ArcFace), sometimes combined with anomaly-detection models (Isolation Forest) on behavioural metadata.[28][29]
- Limitations: Require controlled camera placement and explicit biometric enrollment, raise strong privacy and legal concerns, and are fragile under low-resolution CCTV.

1.3 Object detection and tracking algorithms in surveillance settings

Task	Common algorithms	Typical performance characteristics
Person detection in CCTV	YOLOv5/6/7/8, YOLOX, RT-DETR	Real-time at 640×640 on modern GPUs; accuracy degrades on low-resolution or oblique views.[12][32][^33]
Multi-object tracking	DeepSORT, ByteTrack, OC-SORT, FeatureSORT	Combine detector with Kalman filtering + ReID features; handle moderate occlusions with reduced ID switches.[34][10][11][35][^36]
Person re-identification	Dedicated ReID networks; YOLO-ReID hybrids (e.g., YOLORe-IDNet)	Allow cross-camera association even with partial occlusions, F1≈ 0.79 in campus-style datasets.[^37]
Occlusion-robust tracking	PD-SORT, depth-aware MOT	Use pseudo-depth cues to maintain identity during heavy occlusion events.[^38]

These algorithms are mature enough for real-time surveillance in retail and smart-city deployments, but their performance strongly depends on camera resolution, angle, and scene complexity.

1.4 Success rates and points of failure

Empirical metrics from surveillance-oriented studies suggest that YOLOv8-class detectors can achieve human-detection accuracy (precision/recall) around 75–90% in favourable conditions, but performance drops significantly under poor illumination or extreme angles. DeepSORT/FeatureSORT-style trackers achieve high Multiple Object Tracking Accuracy (MOTA ≈ 80%) on benchmark datasets with moderate occlusion, yet identity switches rise as occlusion and crowd density increase.[11][32][35][39][^10]

Common failure modes relevant to polling stations:

- **Occlusion:** People standing close to booths or in crowds hide each other and the critical area, making it difficult to detect individual entries or gestures.
- **Low resolution / compression:** Many election-grade cameras are configured at low bit-rates, harming small-object detection (e.g., ballots, hands).[^11]
- **Viewpoint issues:** Ceiling-mounted or shallow-angle cameras often cannot see into the booth, by design, which is legally required but limits direct verification of ballot marking.
- **Domain shift:** Training on generic datasets (COCO, MOT) yields sub-optimal performance on Armenian polling-station interiors unless the models are fine-tuned on local data.

2. Alignment with Armenian Electoral Law

2.1 Legal baseline: secrecy and video recording

Armenia's Electoral Code (Constitutional Law, English translation as of May 2018) defines core principles including free suffrage and secrecy of the vote.[^9]

Key provisions:

- **Free suffrage (Article 4):** Elections must be free and voluntary; nobody may force a voter to support or oppose a candidate or to participate or abstain.[^9]
- **Secrecy of voting (Article 6):** Voting is by secret ballot; secrecy is not only a right but also a responsibility of the elector, and any control over the free expression of the elector's will is prohibited.[^9]
- **Publicity and video recording (Article 8(11)–(11.1)):** Proxies, observers, media may photograph and videotape the voting process without violating secrecy, and the Central Electoral Commission (CEC) must contract a specialised organisation to record and live webcast the voting process and summarisation from all precincts using fixed cameras, ensuring that the ballot box, polling station and entrance (within 50 m), registration, and summarisation are in the field of view, while respecting secrecy of voting.[^9]

Thus, Armenia already legally permits and organises extensive video capture at polling stations, explicitly conditioned on non-violation of secrecy.

2.2 Assistance to voters and presence in the booth

The Code and related Venice Commission materials specify the rules for assistance:[40][41][42][9]

- A voter who is **unable to fill in the ballot paper on their own** may, after informing the chairperson, invite another person into the voting booth, who must not be a commission member, proxy, observer, media representative, or visitor.[43][40]
- Traditionally, **one helper may assist only one voter**, and the presence of any other person in the booth while marking the ballot is prohibited.[44][43][^40]
- Proposed amendments discussed in Venice Commission opinions have considered allowing two commission members or proxies to accompany a voter with disabilities so they can observe each other, but the core logic remains that multiple persons in the booth is exceptional and tightly controlled.[41][42]

The Criminal Code separately penalises violations of secrecy, including forcing a voter to reveal their vote, entering the booth without entitlement, or examining the ballot to reveal the vote.[^45]

2.3 Typical violations observed in Armenian practice

Observation missions in Armenia have documented recurring patterns of violations related to secrecy and crowding:

- In the 2017 Yerevan City Council election, independent observers recorded cases where, in 46% of sampled stations, **more than one person entered the voting booth** and, in 77%,

ballots were filled or envelopes packed outside the booth.[^26]

- In snap parliamentary elections, the “Independent Observer” alliance reported that in around 31.5% of observed stations, more than one person entered the booth, and in 30% there was propaganda near polling stations.[^23]
- Other reports describe mirrors or improper booth placement that risk exposing how voters mark ballots.[^46]

These documented behaviours map directly to visual patterns detectable (at least partially) by computer vision: number of people entering booths, people holding ballots outside, observers standing close to booths, and presence of crowds or agitators inside the polling room.

2.4 Mapping CV capabilities to legal violations

The table below maps high-level computer-vision capabilities to specific legal issues in the Armenian context.

CV capability	Observable pattern	Potential legal issue under Armenian law	Notes/constraints
Person detection + region-of-interest (ROI) counters	Number of people entering a delimited “booth ROI” within a time window	Unauthorised presence or multiple persons in the booth; potential breach of secrecy and improper assistance (Electoral Code rules on single-person booth, Criminal Code Article 154).[9][45][40][23][^26]	Cannot see the marked ballot; system should only flag “2+ persons simultaneously in booth” as a risk event, not infer vote choice.
Pose estimation / fine-grained body keypoints	Person pointing at another’s ballot or physically marking ballot for them; standing very close behind a voter at the booth entrance	Coercion or undue influence; improper assistance where helper effectively controls the vote; violation of secrecy and free suffrage.[9][26]	Pose models may struggle in cramped booths; classification accuracy must be treated as indicative rather than conclusive evidence.
Object detection (ballot, envelope, phone)	Ballots being filled or envelopes prepared outside the booth; voters photographing ballots; use of mobile phones near the booth	Breach of secrecy (filling ballots outside booth, ballot photography), potential vote-buying control schemes, prohibited equipment.[9][26][^45]	Requires models trained to detect ballots and phones at low resolution; error rates will be non-trivial.
Crowd counting and density estimation	High density near booths, commission table, or entrance; many non-voters lingering	Unauthorised crowding in the polling room or within 50 m, potential campaigning in vicinity, intimidation of voters.[9][23][^26]	Crowd thresholds must be tuned to each station size; legal definition of “crowding” is qualitative, so AI should supply metrics, not definitive judgments.

Person Re-ID (non-biometric clothing/appearance features)	Same individual entering the polling room or booth multiple times; same escort accompanying many voters	Potential multiple voting or escorting many voters contrary to assistance rules.[40][41][^23]	To respect secrecy and data-protection norms, ReID should avoid storing stable biometric templates; clothing-based embeddings and short-term session-level IDs are safer.
Zone-based tracking and dwell-time analysis	Long dwell time of unauthorised persons near commission table, ballot box, or registration area	Risk of undue influence, interference with commission, or tampering around the ballot box.[^9]	Useful for alerting observers; human review remains essential to interpret context.

2.5 Legal-tech design constraints

Given Article 6 and the explicit requirement to respect secrecy when cameras record and webcast polling-station activities, any AI layer must adhere to key constraints:[47][9]

- **No inference of vote choice:** Models must never attempt to read or infer marks on ballots (e.g., by zooming into the booth) or to predict party preference from visual cues.
- **Camera placement discipline:** Cameras should be positioned to see booth entrances, ballot box, and registration tables, not the interior of the booth where the mark is made, replicating CEC's existing guidelines.[^9]
- **Minimal biometric processing:** Facial recognition of voters for fraud detection is highly sensitive and likely incompatible with secrecy and proportionality principles; if used at all, it should be limited to post-hoc forensic analysis under strict legal authorisation, not routine live monitoring.[29][48] [27][28]
- **Transparency and oversight:** Use of AI must be publicly documented; observers and parties should have access to system specifications, alert logs, and the ability to request stored video consistent with Article 8's rules for access to recordings.[^9]

Within these constraints, pattern-based visual analytics for risk detection and queue/crowd management appear compatible with the Electoral Code's existing framework, provided the system is carefully specified in secondary legislation or CEC decisions.

3. Technical Architecture and Constraints

3.1 Scale, throughput, and high-level architecture

Assumptions for Armenia's 2026 parliamentary elections:

- About **1800 polling stations**, each with at least two fixed cameras: one covering the entrance/queue area, one covering the interior (ballot box, booths, commission table).[24][9]
- Cameras output RTSP/HLS streams at 1080p or 720p, 15–25 fps; video is already centrally recorded for legal reasons (per Article 8(11.1)).[^9]

A scalable architecture for AI analytics on top of this infrastructure can be structured as follows:

1. **Ingestion layer:**
   - RTSP/HLS ingest via GStreamer or FFmpeg into a message/streaming bus (e.g., Kafka, Pravega) or directly into DeepStream pipelines.[49][50]
   - Time synchronisation and metadata tagging (station ID, camera type, region).
2. **Pre-processing / transcoding:**
   - Down-scale for analytics (e.g., 720p → 416p) while maintaining full-quality archival recording.
   - Optional edge ROIDet to pre-crop regions of interest and reduce bandwidth, as in the DeepStream "ROIDet" research that achieved up to 54% bandwidth savings with minimal accuracy loss.[^51]
3. **Inference and tracking layer:**
   - GPU-accelerated pipelines (NVIDIA DeepStream, OpenVINO Model Server, or custom TensorRT/ONNXRuntime services) running:
     - Person and object detection models (YOLOv8/YOLOv9-s/m, RT-DETR or optimised small-object variants like SOD-YOLOv8).[12][52] [^14]
     - Multi-object tracking (DeepSORT/ByteTrack/FeatureSORT) with appearance embeddings and basic ReID.[34][37] [35][10]
     - Optional specialised heads for queue length, crowd density, and simple action recognition (e.g., entering booth, standing too close).
4. **Event extraction and rule engine:**
   - Per-station micro-services summarising low-level detections into higher-level events: "two persons in booth ROI for > X seconds", "same escort ID enters booth with N different voters", "queue length > threshold", "crowd density near booth above Y for Z minutes".
   - Configurable rules, thresholds, and schedules (e.g., lower tolerance during peak hours).
5. **Central monitoring and alerting:**
   - Web dashboard at central and regional command centres, showing per-station status, live thumbnails, and alert feed.
   - Integration with messaging channels for rapid-response teams (Signal/Matrix/secure VoIP) and with case-management tools used by legal NGOs.
6. **Storage and audit:**
   - Long-term storage of raw video (as already mandated) and of structured event logs and alert clip snippets for later investigation and court use.[^9]

This architecture can be deployed either in a national data centre or in public cloud regions that satisfy Armenia's data-sovereignty and connectivity constraints, with VPNs or SD-WAN to connect municipal networks.

3.2 Choice of CV models for low-resolution, angled cameras

Detection models

- **YOLOv8/YOLOv9 small/medium:** These provide strong accuracy–speed trade-offs and have been successfully deployed in intrusion and surveillance scenarios, with near real-time performance and average confidence around 76–90% for human detection in varied conditions.[32][53] [^39]
- **RT-DETR and similar real-time transformers:** DeepStream performance benchmarks show RT-DETR at 640×640 FP16 reaching around 153–160 FPS per high-end GPU in lab conditions, suggesting that tens of 5–10 fps streams can be supported when batched.[^12]
- **Small-object-enhanced variants (SOD-YOLOv8):** Designed specifically to improve detection of small, distant objects (e.g., people or ballots in wide-angle scenes), using improved feature pyramids and attention mechanisms.[^52]

Given polling-station camera constraints (often ceiling-mounted, wide FOV), YOLOv8-s or a SOD-YOLOv8-style model fine-tuned on Armenian polling-station imagery is a pragmatic choice.

Tracking and Re-ID

- **DeepSORT and derivatives (FeatureSORT):** Combine a detector with Kalman filtering and appearance embeddings from a ReID network to maintain identities across frames and short occlusions, explicitly aimed at handling surveillance scenarios.[35][36] [34][10] [^11]
- **YOLORe-IDNet:** Integrates YOLOv5 detection with a ReID module to track suspects across multiple cameras, achieving F1 ≈ 0.79 and IOU ≈ 0.59 on multi-camera datasets at real-time speeds.[^37]

For Armenia's use case, DeepSORT/FeatureSORT with a lightweight clothing-oriented ReID model (colour/style embeddings instead of face) is recommended to balance privacy and performance.

Occlusion handling

- Newer trackers like PD-SORT use pseudo-depth cues and depth-volume IoU to maintain associations in heavy occlusions, improving MOT metrics in crowded scenes.[^38]
- Even with these techniques, severe occlusion (many people near the booth) will degrade accuracy; the system must express uncertainty and rely on escalation to human review rather than hard claims.

3.3 Cloud-infrastructure sizing and latency

Exact sizing depends on chosen models, frame rates, and whether analytics run on every frame or sampled frames, but public benchmarks and vendor guidance allow ballpark estimates.

Per-GPU stream capacity

- NVIDIA DeepStream documentation and community benchmarks suggest that a T4-class GPU can handle roughly 13 streams of 1080p 30 fps YOLOv4 object detection in real time; 100 such streams may require around 8 T4 GPUs.[^54]
- Jetson-class devices running YOLOv8 can support around 4–6 streams (Orin Nano) and 16–18 streams (Orin NX 16 GB) at capacity, indicating that modern datacentre GPUs (A10, L4) can handle tens of streams each when optimised and when per-stream fps is reduced. [^14]
- A recent practitioner report using two NVIDIA A10 GPUs (31.2 TFLOPS each) found that 62 streams (32 at 1080p, 30 at 4K) running YOLOv8-medium + LPRNet at 30 fps vastly exceeded available compute (required ≈ 172 TFLOPS vs 62 TFLOPS), underscoring that naïve full-frame, full-fps inference at high resolution is not feasible without sub-sampling and INT8 optimisation.[^55]

Implication for Armenia (1 800 stations):

- If analytics are run at 8 fps on 720p streams and models are quantised to FP16/INT8, a single A10-class GPU could likely support 20–30 streams (conservative), implying on the order of 60–90 A10-equivalent GPUs for 1 800 streams, plus CPU and storage resources. [54][55] [^12]
- Alternatively, a hierarchical approach with **edge inference** (small nodes per marz or municipality) and central aggregation could reduce backbone bandwidth and central GPU demand.

Network and latency

- A forum benchmark indicates that 100 1080p H.264 streams consume around 400 Mbps of network bandwidth, within a 1 Gbit/s link's capacity.[^54]
- Latency for RTSP access plus decoding and buffering can contribute 500–1 500 ms, with total delays of 700–2 000 ms typical for 30 fps RTSP analytics pipelines.[^13]

For a central "alert system", 1–5 seconds of end-to-end latency is acceptable, as rapid-response teams cannot physically reach a station instantly; thus, the analytics pipeline can safely trade some latency for robustness.

3.4 Handling partial occlusions and camera limitations

Mitigating occlusions and imperfect viewpoints requires both algorithmic and operational strategies:

- **Multi-camera fusion per station:** Where feasible, installing two or three cameras per station with overlapping views (entrance, booths, ballot box) allows cross-view tracking and partial occlusion recovery, using multi-camera MOT techniques similar to those used in pedestrian-tracking research.[8][11]
- **Zone-based rules instead of individual-level certainty:** Rather than trying to determine exactly who did what inside the booth, define small ROIs (booth entrance area, immediate surroundings) and count how many distinct persons are present; even if individual tracks are occasionally lost, aggregated features (max persons simultaneously in ROI, average dwell time) are robust to some occlusion.

- **Feature-rich tracking:** Incorporating colour, clothing style, and direction features (as in FeatureSORT) improves association across short occlusions compared with IoU-only tracking.[56][35]
- **Careful camera placement and calibration:** Pre-election, each station's cameras should be calibrated to ensure that:
  - Booth interiors are not visible (for secrecy) but booth entrances and the surrounding floor are fully in frame.
  - The ballot box and commission table are visible with sufficient pixel density to detect people and approximate gestures.
  - Over-exposure from windows or mixed lighting is minimised.

Despite these mitigations, there will be blind spots and cases where the AI cannot reliably interpret events (e.g., if a group briefly crowds the booth entrance); the system must expose confidence scores and allow operators to view the full video context before classifying an event as a violation.

3.5 Software stack recommendations

A pragmatic, open-standards-oriented stack could include:

- **Ingestion & orchestration:**
  - GStreamer/FFmpeg for RTSP/HLS capture; Kubernetes for scaling per-station analytics pods.[50][49]
  - Kafka or Pravega for decoupled video and metadata streams.[^49]
- **Inference:**
  - NVIDIA DeepStream SDK with TensorRT-optimised YOLOv8/RT-DETR models on datacentre GPUs, or Intel OpenVINO Model Server for CPU/VPU-accelerated inference.[50][14] [^12]
- **Tracking and event logic:**
  - DeepSORT/FeatureSORT implementations integrated into the DeepStream or OpenVINO pipeline; configurable rule engine (e.g., Flink, custom micro-service) to translate trajectories into events.[10][56] [^35]
- **Dashboard & operations:**
  - Web dashboards built with standard web frameworks, with role-based access for CEC, civil-society coalitions, and party representatives; integration of alert feeds with incident-management tools used by NGOs.

This stack is compatible with commodity x86 servers equipped with A10/L4 GPUs in a central data centre or private cloud.

4. AI vs. Human Observers: Effectiveness and Synergy

4.1 Cost comparison (high-level)

Human observers

- Armenia had around 2 008 polling stations in recent parliamentary elections; many domestic observer missions fielded several hundred observers, often covering each station for part of the day, supported by mobile teams.[57][23]
- If one assumes 1 full-time observer per station for the 14–16 hour election day plus some reserves, this implies roughly 1 800–2 000 observers.
- Average monthly salary in Armenia was about 279 000 drams in late 2024; daily wages for comparable skilled work might approximate 10 000–15 000 drams, while the national minimum wage in 2023–2025 is around 75 000 drams per month, increasing toward 85 000 drams by 2026.[58][59] [60][61] [^62]

Very roughly, hiring 1 800 observers at 15 000 drams for a day would cost on the order of 27 million drams per election day, not counting training, coordination staff, transport, and legal support.

AI system

- Capital costs include GPU servers (tens of A10/L4-class GPUs), networking, storage, and development of custom analytics and dashboards; operational costs include electricity, maintenance, and a small monitoring staff.
- While there is no precise public price reference, running a 60–90-GPU cluster plus supporting infrastructure for an election cycle is likely comparable to or higher than hiring observers for a single day, but the hardware can be reused across elections and for other civic-tech purposes.
- The **marginal cost** of monitoring more stations with AI is low once infrastructure is in place, while human observer costs scale linearly with the number of stations and hours.

Given these rough figures, AI monitoring should be viewed not as a direct cost substitute for observers, but as an infrastructure investment that amortises across multiple elections and provides continuous capabilities (archival search, training datasets, etc.).

4.2 Comparative effectiveness

Strengths of human observers:

- Understand context, intent, and complex interactions that are visually ambiguous (e.g., genuine assistance vs. coercion, subtle campaigning, psychological pressure).
- Can file complaints, intervene, and deter fraud by their presence.
- Can observe non-visual aspects (instructions being given, tone of voice, conversations outside the camera's field of view).

Strengths of AI monitoring:

- Continuous coverage of all monitored stations simultaneously, including at night or low-attention periods where observers may be tired.
- Capacity to detect statistical patterns (e.g., repeated escorts, unusual booth occupancy patterns) that are impossible for individual observers to track mentally across many hours.
- Objective, timestamped event logs and clips that can support investigations and court cases.

Limitations of AI:

- Susceptible to false positives/negatives due to occlusion, lighting, or camera shift.
- Cannot interpret legal nuances and must not autonomously label behaviour as "fraud" or "crime" without human review.
- Risk of chilling effects or public backlash if perceived as surveillance that infringes secrecy or privacy.

Therefore, AI should be positioned as a **force multiplier** rather than a replacement for observers.

4.3 Operational strategy for AI–human synergy

A realistic operational model for Armenia's 2026 elections could follow these principles:

1. **Central AI monitoring hub:**
   - Located in Yerevan, mirrored by regional centres; staffed by a small team of technical operators and legal coordinators.
   - Receives event streams from all polling-station cameras via the AI pipeline.
2. **Tiered alerting:**
   - Level 1 (informational): queue length, crowd density, camera offline, suspicious but low-confidence patterns (e.g., potential dual presence in booth) – displayed on dashboards and used to focus attention.
   - Level 2 (priority): multiple Level 1 events at the same station (e.g., repeated booth crowding, many escorts) – triggers immediate video review by human operators.
   - Level 3 (critical): high-confidence events (e.g., clear multiple persons in booth for prolonged periods, obvious ballot handling outside booth) – escalated to legal teams and mobile observers.
3. **Integration with observer networks:**
   - Domestic observer coalitions and party proxies receive read-only access to parts of the alert dashboard and can subscribe to alerts for specific marzes or stations.
   - When an alert is confirmed by central reviewers, the coalition dispatches nearby mobile observers or lawyers to the station, or contacts the territorial commission as appropriate.
4. **Legal workflows and evidence management:**

- Each confirmed alert generates an incident file with video clips, AI metadata, and human annotations; this file feeds into the complaint-submission workflow to the CEC or courts.
- Access controls prevent misuse of video data (no face recognition, time-limited access, logging of who viewed which incident).

5. **Public communication and transparency:**
   - Before the election, the CEC and civil-society partners publish clear documentation of what the AI system does and does not do (no inference of vote choice, no mass facial recognition), and how alerts are handled.
   - After the election, aggregated statistics on alerts and confirmed violations are reported, including known false positives, to build trust and enable iterative improvements.

4.4 Practical feasibility assessment

Technical feasibility:
Given existing precedents in India and Armenia for webcasting and AI video analytics at tens of thousands of sites, plus mature open-source and commercial stacks for multi-camera surveillance, the technical feasibility of monitoring 1800 Armenian polling stations in near real time is high, provided sufficient budget, connectivity, and 6–12 months of dedicated development and field testing.[16][17] [6][2] [7][51] [1][19] [24][12]

Legal feasibility:
The Electoral Code already foresees comprehensive video recording and webcasting, and observation missions have long used video evidence to document violations; augmenting this with AI analytics focused on crowd patterns and procedural anomalies is legally plausible if safeguards are built into secondary regulations and system design. Automated biometric identification of voters, however, would likely conflict with secrecy and data-protection principles and should be avoided or strictly confined.[23][26] [^9]

Operational feasibility:
Building an integrated central AI hub, rule engine, and dashboards, and training legal and observer teams to use them, is a substantial but manageable project, especially if development starts in 2025 with pilots in by-elections or mock exercises.

Risk profile:
Main risks include technical failures (network outages, model mis-calibration), misinterpretation of AI outputs, political controversy around “surveillance”, and adversarial adaptation (fraudsters exploiting blind spots). These risks can be reduced by:

- Ensuring redundancy in connectivity and recording.
- Keeping AI outputs strictly advisory, requiring human review for any formal complaint.
- Involving a broad coalition of stakeholders (CEC, parties, NGOs, international observers) in system design and governance.

Under these conditions, a nationwide AI-assisted monitoring system for Armenia's 2026 parliamentary elections is **practically feasible and strategically valuable**, provided expectations are realistic: the system will not eliminate fraud, but it can substantially increase the probability that violations are detected, recorded, and acted upon.

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